2.09 Binomial Series

Worked Solutions 1–23

1 a)

$$\begin{aligned}(1+x)^{\frac{1}{4}} &\approx 1+\tfrac{1}{4}x+\tfrac{1}{2}(\tfrac{1}{4})(-\tfrac{3}{4})x^2+\tfrac{1}{6}(\tfrac{1}{4})(-\tfrac{3}{4})(-\tfrac{7}{4})x^3+\ldots\\ &\approx 1+\tfrac{1}{4}x-\tfrac{3}{32}x^2+\tfrac{7}{128}x^3+\ldots\end{aligned}$$

$|x|<1$.

1 b)

$$\begin{aligned}(1+2x)^{-1} &\approx 1-2x+\tfrac{1}{2}(-1)(-2)(2x)^2+\tfrac{1}{6}(-1)(-2)(-3)(2x)^3+\ldots\\ &\approx 1-2x+4x^2-8x^3+\ldots\end{aligned}$$

$|2x|<1 \Rightarrow |x|<\frac{1}{2}$.

1 c)

$$\begin{aligned}(1-3x)^{\frac{1}{2}} &\approx 1+\tfrac{1}{2}(-3x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(-3x)^2+\tfrac{1}{6}(\tfrac{1}{2})(-\tfrac{1}{2})(-\tfrac{3}{2})(-3x)^3+\ldots\\ &\approx 1-\tfrac{3}{2}x-\tfrac{9}{8}x^2-\tfrac{27}{16}x^3+\ldots\end{aligned}$$

$|3x|<1 \Rightarrow |x|<\frac{1}{3}$.

1 d)

$(2+x)^{-3}=2^{-3}(1+\frac{1}{2}x)^{-3}$.

$$\begin{aligned}&\approx \tfrac{1}{8}[1-3(\tfrac{1}{2}x)+\tfrac{1}{2}(-3)(-4)(\tfrac{1}{2}x)^2+\tfrac{1}{6}(-3)(-4)(-5)(\tfrac{1}{2}x)^3+\ldots]\\ &\approx \tfrac{1}{8}-\tfrac{3}{16}x+\tfrac{3}{16}x^2-\tfrac{5}{32}x^3+\ldots\end{aligned}$$

$|\frac{1}{2}x|<1 \Rightarrow |x|<2$.

1 e)

$(9+3x)^{\frac{1}{2}}=9^{\frac{1}{2}}(1+\frac{1}{3}x)^{\frac{1}{2}}$.

$$\begin{aligned}&\approx 3[1+\tfrac{1}{2}(\tfrac{1}{3}x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(\tfrac{1}{3}x)^2+\tfrac{1}{6}(\tfrac{1}{2})(-\tfrac{1}{2})(-\tfrac{3}{2})(\tfrac{1}{3}x)^3+\ldots]\\ &\approx 3+\tfrac{1}{2}x-\tfrac{1}{24}x^2+\tfrac{1}{144}x^3+\ldots\end{aligned}$$

$|\frac{1}{3}x|<1 \Rightarrow |x|<3$.

1 f)

$(16-36x)^{\frac{1}{2}}=16^{\frac{1}{2}}(1-\frac{9}{4}x)^{\frac{1}{2}}$.

$$\begin{aligned}&\approx 4[1+\tfrac{1}{2}(-\tfrac{9}{4}x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(-\tfrac{9}{4}x)^2+\tfrac{1}{6}(\tfrac{1}{2})(-\tfrac{1}{2})(-\tfrac{3}{2})(-\tfrac{9}{4}x)^3+\ldots]\\ &\approx 4-\tfrac{9}{2}x-\tfrac{81}{32}x^2-\tfrac{729}{256}x^3+\ldots\end{aligned}$$

$|\frac{9}{4}x|<1 \Rightarrow |x|<\frac{4}{9}$.

1 g)

$(8-2x)^{\frac{1}{3}} = 8^{\frac{1}{3}}(1-\frac{1}{4}x)^{\frac{1}{3}}$.

$$\begin{aligned}&\approx 2[1+\tfrac{1}{3}(-\tfrac{1}{4}x)+\tfrac{1}{2}(\tfrac{1}{3})(-\tfrac{2}{3})(-\tfrac{1}{4}x)^2+\tfrac{1}{6}(\tfrac{1}{3})(-\tfrac{2}{3})(-\tfrac{5}{3})(-\tfrac{1}{4}x)^3+\ldots]\\&\approx 2-\tfrac{1}{6}x-\tfrac{1}{72}x^2-\tfrac{5}{2592}x^3+\ldots\end{aligned}$$

$|\frac{1}{4}x|<1 \Rightarrow |x|<4$.

1 h)

$(3+2x)^{-\frac{1}{2}} = 3^{-\frac{1}{2}}(1+\frac{2}{3}x)^{-\frac{1}{2}}$.

$$\begin{aligned}&\approx \tfrac{1}{\sqrt{3}}[1-\tfrac{1}{2}(\tfrac{2}{3}x)+\tfrac{1}{2}(-\tfrac{1}{2})(-\tfrac{3}{2})(\tfrac{2}{3}x)^2+\tfrac{1}{6}(-\tfrac{1}{2})(-\tfrac{3}{2})(-\tfrac{5}{2})(\tfrac{2}{3}x)^3+\ldots]\\&\approx \tfrac{1}{\sqrt{3}}-\tfrac{1}{3\sqrt{3}}x+\tfrac{1}{6\sqrt{3}}x^2-\tfrac{5}{54\sqrt{3}}x^3+\ldots\\&= \tfrac{\sqrt{3}}{3}-\tfrac{\sqrt{3}}{9}x+\tfrac{\sqrt{3}}{18}x^2-\tfrac{5\sqrt{3}}{162}x^3+\ldots\end{aligned}$$

$|\frac{2}{3}x|<1 \Rightarrow |x|<\frac{3}{2}$.

2 a)

$\frac{1}{4!}\times\frac{1}{4}(-\frac{3}{4})(-\frac{7}{4})(-\frac{11}{4})(3x)^4 = -\frac{6237}{2048}x^4$. So the coefficient is $-\frac{6237}{2048}$.

2 b)

$\frac{1}{4!}\times\frac{2}{3}(-\frac{1}{3})(-\frac{4}{3})(-\frac{7}{3})(-2x)^4 = -\frac{112}{243}x^4$. So the coefficient is $-\frac{112}{243}$.

2 c)

$\frac{1}{4!}\times -\frac{3}{4}(-\frac{7}{4})(-\frac{11}{4})(-\frac{15}{4})(\frac{1}{2}x)^4 = \frac{1155}{32768}x^4$. So the coefficient is $\frac{1155}{32768}$.

3 a)

$(9+x)^{-\frac{1}{2}} = 9^{-\frac{1}{2}}(1+\frac{1}{9}x)^{-\frac{1}{2}}$.

$$\begin{aligned}&\approx \tfrac{1}{3}[1-\tfrac{1}{2}(\tfrac{1}{9}x)+\tfrac{1}{2}(-\tfrac{1}{2})(-\tfrac{3}{2})(\tfrac{1}{9}x)^2+\ldots]\\&\approx \tfrac{1}{3}-\tfrac{1}{54}x+\tfrac{1}{648}x^2+\ldots\end{aligned}$$

3 b)

$$\begin{aligned}(9-x^2)^{-\frac{1}{2}} &\approx \tfrac{1}{3}-\tfrac{1}{54}(-x^2)+\tfrac{1}{648}(-x^2)^2+\ldots\\&\approx \tfrac{1}{3}+\tfrac{1}{54}x^2+\tfrac{1}{648}x^4+\ldots\end{aligned}$$

4 a)

$$\begin{aligned}(1+x)^{-\frac{1}{2}} &\approx 1-\tfrac{1}{2}x+\tfrac{1}{2}(-\tfrac{1}{2})(-\tfrac{3}{2})x^2+\ldots\\&\approx 1-\tfrac{1}{2}x+\tfrac{3}{8}x^2+\ldots\end{aligned}$$

4 b)

$$\begin{aligned}(1+\tfrac{2}{5}x)^{-\frac{1}{2}} &\approx 1-\tfrac{1}{2}(\tfrac{2}{5}x)+\tfrac{3}{8}(\tfrac{2}{5}x)^2+\ldots\\ &\approx 1-\tfrac{1}{5}x+\tfrac{3}{50}x^2+\ldots\end{aligned}$$

$$\begin{aligned}\sqrt{10}(5+2x)^{-\frac{1}{2}} &= \sqrt{10}\times\tfrac{1}{\sqrt{5}}(1+\tfrac{2}{5}x)^{-\frac{1}{2}}\\ &\approx \sqrt{2}[1-\tfrac{1}{5}x+\tfrac{3}{50}x^2+\ldots]\\ &\approx \sqrt{2}-\tfrac{\sqrt{2}}{5}x+\tfrac{3\sqrt{2}}{50}x^2+\ldots\end{aligned}$$

5 a)

$(9-x)^{\frac{1}{2}} = 9^{\frac{1}{2}}(1-\frac{1}{9}x)^{\frac{1}{2}}$.

$$\begin{aligned}&\approx 3[1+\tfrac{1}{2}(-\tfrac{1}{9}x)+\tfrac{\frac{1}{2}(-\frac{1}{2})}{2}(-\tfrac{1}{9}x)^2+\ldots]\\ &\approx 3-\tfrac{1}{6}x-\tfrac{1}{216}x^2+\ldots\end{aligned}$$

5 b)

$|\frac{1}{9}x|<1 \Rightarrow |x|<9$ so can use $x=1$.

$$\sqrt{8}=\sqrt{9-1}\approx 3-\tfrac{1}{6}-\tfrac{1}{216}+\ldots\approx\tfrac{611}{216}$$

6 a)

$$\begin{aligned}(1+5x)^{\frac{1}{3}} &\approx 1+\tfrac{1}{3}(5x)+\tfrac{1}{2}(\tfrac{1}{3})(-\tfrac{2}{3})(5x)^2+\tfrac{1}{6}(\tfrac{1}{3})(-\tfrac{2}{3})(-\tfrac{5}{3})(5x)^3+\ldots\\ &\approx 1+\tfrac{5}{3}x-\tfrac{25}{9}x^2+\tfrac{625}{81}x^3+\ldots\end{aligned}$$

6 b)

$1+5x=1.1125 \Rightarrow x=\frac{9}{400}$. $|\frac{1}{5}x|<1 \Rightarrow |x|<5$ so can use $x=\frac{9}{400}$.

$$\begin{aligned}\sqrt[3]{1.1125} &\approx 1+\tfrac{5}{3}(\tfrac{9}{400})-\tfrac{25}{9}(\tfrac{9}{400})^2+\tfrac{625}{81}(\tfrac{9}{400})^3+\ldots\\ &\approx \tfrac{21221}{20480}\approx 1.0362\end{aligned}$$

7)

$(2-3x)^{\frac{1}{2}} = 2^{\frac{1}{2}}(1-\frac{3}{2}x)^{\frac{1}{2}}$.

$$\begin{aligned}&\approx \sqrt{2}[1+\tfrac{1}{2}(-\tfrac{3}{2}x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(-\tfrac{3}{2}x)^2+\ldots]\\ &\approx \sqrt{2}-\tfrac{3\sqrt{2}}{4}x-\tfrac{9\sqrt{2}}{32}x^2+\ldots\end{aligned}$$

$|\frac{3}{2}x|<1 \Rightarrow |x|<\frac{2}{3}$ so $x=\frac{1}{10}$ is valid.

$$\begin{aligned}\sqrt{\tfrac{17}{10}} &\approx \sqrt{2}-\tfrac{3\sqrt{2}}{4}\times\tfrac{1}{10}-\tfrac{9\sqrt{2}}{32}\times\tfrac{1}{100}+\ldots=\tfrac{2951}{3200}\sqrt{2}\\ \sqrt{170} &= 10\times\sqrt{\tfrac{17}{10}}\approx 10\times\tfrac{2951}{3200}\sqrt{2}=\tfrac{2951}{320}\sqrt{2}\end{aligned}$$

8 a)

$$\begin{aligned}(1+3x)^{\frac{1}{2}} &\approx 1+\tfrac{1}{2}(3x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(3x)^2+\ldots\\ &\approx 1+\tfrac{3}{2}x-\tfrac{9}{8}x^2+\ldots\end{aligned}$$

$|3x|<1 \Rightarrow |x|<\frac{1}{3}$.

$$\begin{aligned}(1+x)^{-\frac{1}{2}} &\approx 1-\tfrac{1}{2}x+\tfrac{1}{2}(-\tfrac{1}{2})(-\tfrac{3}{2})x^2+\ldots\\ &\approx 1-\tfrac{1}{2}x+\tfrac{3}{8}x^2+\ldots\end{aligned}$$

$|x|<1$.

8 b)

$$\begin{aligned}\sqrt{\frac{1+3x}{1+x}} &= (1+3x)^{\frac{1}{2}}(1+x)^{-\frac{1}{2}}\\ &= (1+\tfrac{3}{2}x-\tfrac{9}{8}x^2+\ldots)(1-\tfrac{1}{2}x+\tfrac{3}{8}x^2+\ldots)\\ &= 1-\tfrac{1}{2}x+\tfrac{3}{8}x^2+\tfrac{3}{2}x-\tfrac{3}{4}x^2-\tfrac{9}{8}x^2\ldots\\ &= 1+x-\tfrac{3}{2}x^2+\ldots \text{ as required}\end{aligned}$$

8 c)

Expansion valid for $|x|<\frac{1}{3}$.

9 a)

$$\begin{aligned}(1-2x)^{\frac{1}{2}} &\approx 1+\tfrac{1}{2}(-2x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(-2x)^2+\ldots\\ &\approx 1-x-\tfrac{1}{2}x^2+\ldots\end{aligned}$$

9 b)

$\frac{\sqrt{1-2x}}{1+x} = (1-2x)^{\frac{1}{2}}(1+x)^{-1}$.

$$\begin{aligned}(1+x)^{-1} &\approx 1-x+\tfrac{1}{2}(-1)(-2)x^2+\ldots\\ &\approx 1-x+x^2+\ldots\end{aligned}$$

$$\begin{aligned}\text{So } &(1-x-\tfrac{1}{2}x^2+\ldots)(1-x+x^2+\ldots)\\ &= 1-x+x^2-x+x^2-x^3-\tfrac{1}{2}x^2+\tfrac{1}{2}x^3+\ldots\\ &= 1-2x+\tfrac{3}{2}x^2+\ldots \text{ as required}\end{aligned}$$

9 c)

$|x|<1$ and $|2x|<1 \Rightarrow |x|<\frac{1}{2}$. Expansion valid for $|x|<\frac{1}{2}$.

10 a)

$pq = -2$ and $\frac{1}{2}(p)(p-1)q^2 = 6$.

$$\Rightarrow \tfrac{1}{2}p^2q^2 - \tfrac{1}{2}pq^2 = 6$$
$$\Rightarrow 2 + q = 6$$

So $p = -\frac{1}{2}$, $q = 4$.

10 b)

$|4x| < 1 \Rightarrow |x| < \frac{1}{4}$.

11 a)

$(8-x)^{\frac{1}{3}} = 8^{\frac{1}{3}}(1 - \frac{1}{8}x)^{\frac{1}{3}}$.

$$\approx 2[1 - \tfrac{1}{3}(\tfrac{1}{8}x) + \tfrac{1}{2} \times \tfrac{1}{3}(-\tfrac{2}{3})(\tfrac{1}{8}x)^2 + \ldots]$$
$$\approx 2 - \tfrac{1}{12}x - \tfrac{1}{288}x^2 + \ldots \text{ so } k = -\tfrac{1}{288}$$

11 b)

$(8-1)^{\frac{1}{3}} \approx 2 - \frac{1}{12} - \frac{1}{288} + \ldots \approx \frac{551}{288} = 1.913$ to 3 dp.

12 a)

$(3-8x)^{\frac{1}{2}} = 3^{\frac{1}{2}}(1 - \frac{8}{3}x)^{\frac{1}{2}}$.

$$\approx \sqrt{3}[1 - \tfrac{1}{2}(\tfrac{8}{3}x) + \tfrac{1}{2} \times \tfrac{1}{2}(-\tfrac{1}{2})(\tfrac{8}{3}x)^2 + \ldots]$$
$$\approx \sqrt{3}[1 - \tfrac{4}{3}x - \tfrac{8}{9}x^2 + \ldots]$$

$(3 - 8 \times \frac{1}{8})^{\frac{1}{2}} = \sqrt{2}$.

$$(3 - 8 \times \tfrac{1}{8})^{\frac{1}{2}} \approx \sqrt{3}[1 - \tfrac{4}{3} \times \tfrac{1}{8} - \tfrac{8}{9}(\tfrac{1}{8})^2 + \ldots]$$
$$\approx \tfrac{59}{72}\sqrt{3}$$

12 b)

Overestimate since all the terms in the expansion except the first are negative.

13 a)

$(2-x)^{-\frac{1}{2}} = 2^{-\frac{1}{2}}(1 - \frac{1}{2}x)^{-\frac{1}{2}}$.

$$\approx \tfrac{1}{\sqrt{2}}[1 - \tfrac{1}{2}(-\tfrac{1}{2}x) + \tfrac{1}{2} \times (-\tfrac{1}{2})(-\tfrac{3}{2})(-\tfrac{1}{2}x)^2 + \ldots]$$
$$= \tfrac{1}{\sqrt{2}}[1 + \tfrac{1}{4}x + \tfrac{3}{32}x^2 + \ldots]$$
$$= \tfrac{1}{\sqrt{2}} + \tfrac{1}{4\sqrt{2}}x + \tfrac{3}{32\sqrt{2}}x^2 + \ldots$$
$$= \tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{8}x + \tfrac{3\sqrt{2}}{64}x^2 + \ldots$$

13 b)

Let the expansion be $f(x)$. At $x=-3$: $\frac{1}{\sqrt{2+3}}=\frac{1}{\sqrt{5}}$ so $\sqrt{5}=5f(-3)$. At $x=-\frac{6}{5}$: $\frac{1}{\sqrt{2-\frac{6}{5}}}=\frac{\sqrt{5}}{2}$ so $\sqrt{5}=2f(-\frac{6}{5})$. At $x=\frac{1}{5}$: $\frac{1}{\sqrt{2-\frac{1}{5}}}=\frac{\sqrt{5}}{3}$ so $\sqrt{5}=3f(\frac{1}{5})$.

13 c)

Since it is valid for $|\frac{1}{2}x|<1\Rightarrow|x|<2$, $x=-3$ may not be used.

14)

$(7-x)^{-\frac{1}{2}}=7^{-\frac{1}{2}}(1-\frac{1}{7}x)^{-\frac{1}{2}}$.

$$\begin{aligned}&\approx\tfrac{1}{\sqrt{7}}[1-\tfrac{1}{2}(-\tfrac{1}{7}x)+\tfrac{1}{2}\times(-\tfrac{1}{2})(-\tfrac{3}{2})(-\tfrac{1}{7}x)^2+\ldots]\\&=\tfrac{1}{\sqrt{7}}[1+\tfrac{1}{14}x+\tfrac{3}{392}x^2+\ldots]\\&=\tfrac{1}{\sqrt{7}}+\tfrac{1}{14\sqrt{7}}x+\tfrac{3}{392\sqrt{7}}x^2+\ldots\end{aligned}$$

$|\frac{1}{7}x|<1\Rightarrow|x|<7$.

$$(7-4)^{-\frac{1}{2}}=\tfrac{1}{\sqrt{3}}\approx\tfrac{1}{\sqrt{7}}+\tfrac{1}{14\sqrt{7}}\times4+\tfrac{3}{392\sqrt{7}}\times4^2\approx\tfrac{69}{49\sqrt{7}}$$

So $\sqrt{3}\approx\frac{49\sqrt{7}}{69}$.

15)

With each additional term the approximation should get closer to the exact result within the range of validity. $y=1+\frac{1}{2}x^2$ is a positive quadratic graph so this means the second term increases in value and is further away from the graph of $\sqrt{1-x^2}$ than $y=1$.

16)

$(9-x)^{-\frac{1}{2}}=\frac{1}{3}(1-\frac{1}{9}x)^{-\frac{1}{2}}$.

$$\begin{aligned}&\approx\tfrac{1}{3}[1-\tfrac{1}{2}(-\tfrac{1}{9}x)+\tfrac{1}{2}\times(-\tfrac{1}{2})(-\tfrac{3}{2})(-\tfrac{1}{9}x)^2+\ldots]\\&\approx\tfrac{1}{3}[1+\tfrac{1}{18}x+\tfrac{1}{216}x^2+\ldots]\\&\approx\tfrac{1}{3}+\tfrac{1}{54}x+\tfrac{1}{648}x^2+\ldots\end{aligned}$$

$$\begin{aligned}&(a+bx)(\tfrac{1}{3}+\tfrac{1}{54}x+\tfrac{1}{648}x^2-\tfrac{5}{34992}x^3+\ldots)\\&=1-\tfrac{11}{18}x-\tfrac{7}{216}x^2-\tfrac{31}{11664}x^3\end{aligned}$$

$a=3$. $\frac{3}{54}+\frac{1}{3}b=-\frac{11}{18}$ so $b=-2$.

17 a)

$$\begin{aligned}(1+px)^{\frac{1}{2}}&=1+\tfrac{1}{2}px+\tfrac{1}{2}\times\tfrac{1}{2}(-\tfrac{1}{2})p^2x^2+\tfrac{1}{6}\times\tfrac{1}{2}(-\tfrac{1}{2})(-\tfrac{3}{2})p^3x^3+\ldots\\&=1+\tfrac{1}{2}px-\tfrac{1}{8}p^2x^2+\tfrac{1}{16}p^3x^3+\ldots\end{aligned}$$

17 b)

$(1+qx)(1+\frac{1}{2}px-\frac{1}{8}p^2x^2+\frac{1}{16}p^3x^3+\ldots)$. $-4=\frac{1}{2}p+q$ and $-14=-\frac{1}{8}p^2+\frac{1}{2}pq$. $-14=-\frac{1}{8}p^2+\frac{1}{2}p(-4-\frac{1}{2}p)$. $\frac{3}{8}p^2+2p-14=0$ so $p=-\frac{28}{3}$ or $p=4$. And $q=\frac{2}{3}$ or $q=-6$. But considering x^3 term: $-\frac{1}{8}p^2q+\frac{1}{16}p^3=-14$ shows $p=4$, $q=-6$.

18 a)

$$\begin{aligned}(1-x)^{-2} &\approx 1-2(-x)+\tfrac{1}{2}(-2)(-3)(-x)^2+\tfrac{1}{6}(-2)(-3)(-4)(-x)^3+\ldots\\ &\approx 1+2x+3x^2+4x^3+\ldots\end{aligned}$$

18 b)

Since $(b-x)^{-2}=b^{-2}(1-\frac{1}{b}x)^{-2}$

$$\begin{aligned}&\approx \tfrac{1}{b^2}[1+\tfrac{2}{b}x+\tfrac{3}{b}x^2+\tfrac{4}{b}x^3+\ldots]\\ &\approx \tfrac{1}{b^2}+\tfrac{2}{b^3}x+\tfrac{3}{b^3}x^2+\tfrac{4}{b^3}x^3+\ldots\end{aligned}$$

$$2+3x+4x^2+5x^3+\ldots=(a-x)(\tfrac{1}{b^2}+\tfrac{2}{b^3}x+\tfrac{3}{b^3}x^2+\tfrac{4}{b^3}x^3+\ldots)$$

Which gives $\frac{a}{b^2}=2$ and $\frac{2a}{b^3}-\frac{1}{b^2}=3$. $\frac{4}{b}-\frac{1}{b^2}=3\Rightarrow 3b^2-4b+1=0$. With $b=\frac{1}{3}$ or $b=1$. But a and b are integers so $b=1$ and $a=2$. Or

$$\begin{aligned}&2+3x+4x^2+5x^3+\ldots\\ &=1+2x+3x^2+4x^3+\ldots+1+x+x^2+x^3+\ldots\\ &=(1-x)^{-2}+(1-x)^{-1}=\frac{1}{(1-x)^2}+\frac{1}{1-x}\\ &=\frac{1}{(1-x)^2}+\frac{1-x}{(1-x)^2}=\frac{2-x}{(1-x)^2}\end{aligned}$$

So $a=2$ and $b=1$.

19 a)

$\frac{2-5x}{(x-1)(2x-1)}=\frac{A}{x-1}+\frac{B}{2x-1}$. $2-5x=A(2x-1)+B(x-1)$. $2=-A-B$ and $-5=2A+B$ so $A=-3$, $B=1$.

19 b)

$\frac{2-5x}{(x-1)(2x-1)}=-\frac{3}{x-1}+\frac{1}{2x-1}$.

$$\begin{aligned}(1-x)^{-1} &\approx 1-(-x)+\tfrac{1}{2}(-1)(-2)(-x)^2+\ldots\\ &\approx 1+x+x^2+\ldots\\ (-1+2x)^{-1} &= -(1-2x)^{-1}\\ &\approx -[1+2x+(2x)^2+\ldots]\approx -1-2x-4x^2-\ldots\end{aligned}$$

$$\begin{aligned}-\frac{3}{x-1}+\frac{1}{2x-1} &\approx 3(1-x)^{-1}+(-1+2x)^{-1}\\ &\approx 3(1+x+x^2+\ldots)+(-1-2x-4x^2-\ldots)\end{aligned}$$

Hence $\frac{2-5x}{(x-1)(2x-1)}\approx 2+x-x^2-\ldots$.

20 a)

$\frac{11x+5}{(5x+1)(x+3)} = \frac{A}{5x+1} + \frac{B}{x+3}$. $11x + 5 = A(x + 3) + B(5x + 1)$. $11 = A + 5B$. $5 = 3A + B$.
$\frac{11x+5}{(5x+1)(x+3)} = \frac{1}{5x+1} + \frac{2}{x+3}$.

$$\begin{aligned}(1+5x)^{-1} &\approx 1 - 1(5x) + \tfrac{1}{2}(-1)(-2)(5x)^2 + \ldots \\ &\approx 1 - 5x + 25x^2 + \ldots \\ (3+x)^{-1} &= \tfrac{1}{3}(1+\tfrac{1}{3}x)^{-1} \\ &\approx \tfrac{1}{3}[1 - 1(\tfrac{1}{3}x) + \tfrac{1}{2}(-1)(-2)(\tfrac{1}{3}x)^2 + \ldots] \\ &\approx \tfrac{1}{3} - \tfrac{1}{9}x + \tfrac{1}{27}x^2 - \ldots\end{aligned}$$

$$\begin{aligned}\frac{1}{5x+1} + \frac{2}{x+3} &\approx (1+5x)^{-1} + 2(3+x)^{-1} \\ &\approx (1 - 5x + 25x^2 + \ldots) + 2(\tfrac{1}{3} - \tfrac{1}{9}x + \tfrac{1}{27}x^2 - \ldots)\end{aligned}$$

Hence $\frac{11x+5}{(5x+1)(x+3)} \approx \frac{5}{3} - \frac{47}{9}x + \frac{677}{27}x^2 - \ldots$.

20 b)

$|5x| < 1 \Rightarrow |x| < \frac{1}{5}$ and $|\frac{1}{3}x| < 1 \Rightarrow |x| < 3$ so $|x| < \frac{1}{5}$.

21 a)

$\frac{9x^2-5x+2}{(x+1)(3x+1)^2} = \frac{A}{x+1} + \frac{B}{3x+1} + \frac{C}{(3x+1)^2}$. $9x^2 - 5x + 2 = A(3x + 1)^2 + B(3x + 1)(x + 1) + C(x + 1)$.
$9 = 9A + 3B$. $-5 = 6A + 4B + C$. $2 = A + B + C$.

$$\frac{9x^2 - 5x + 2}{(x+1)(3x+1)^2} = \frac{4}{x+1} - \frac{9}{3x+1} + \frac{7}{(3x+1)^2}$$

$$\begin{aligned}(1+x)^{-1} &\approx 1 - 1(x) + \tfrac{1}{2}(-1)(-2)(x)^2 + \ldots \\ &\approx 1 - x + x^2 + \ldots \\ (1+3x)^{-1} &\approx 1 - 3x + 9x^2 + \ldots \\ (1+3x)^{-2} &\approx 1 - 2(3x) + \tfrac{1}{2}(-2)(-3)(3x)^2 + \ldots \approx\end{aligned}$$

$$\begin{aligned}&\frac{4}{x+1} - \frac{9}{3x+1} + \frac{7}{(3x+1)^2} \\ &= 4(x+1)^{-1} - 9(3x+1)^{-1} + 7(3x+1)^{-2} \quad 1 - 6x + 27x^2 + \ldots\end{aligned}$$

Hence

$$\begin{aligned}\frac{9x^2 - 5x + 2}{(x+1)(3x+1)^2} &\approx 4(1 - x + x^2 + \ldots) - 9(1 - 3x + 9x^2 + \ldots) + 7(1 - 6x + 27x^2 + \ldots) \\ &\approx 2 - 19x + 112x^2 - \ldots\end{aligned}$$

21 b)

$|3x| < 1 \Rightarrow |x| < \frac{1}{3}$ and $|x| < 1$ so $|x| < \frac{1}{3}$.

22 a)

$(9-x)^{-\frac{1}{2}} = \frac{1}{3}(1-\frac{1}{9}x)^{-\frac{1}{2}}$.

$$\begin{aligned} &\approx \tfrac{1}{3}[1-\tfrac{1}{2}(-\tfrac{1}{9}x)+\tfrac{1}{2}\times(-\tfrac{1}{2})(-\tfrac{3}{2})(-\tfrac{1}{9}x)^2+\ldots] \\ &\approx \tfrac{1}{3}[1+\tfrac{1}{18}x+\tfrac{1}{216}x^2+\ldots] \\ &\approx \tfrac{1}{3}+\tfrac{1}{54}x+\tfrac{1}{648}x^2+\ldots \end{aligned}$$

22 b)

$(9-x^2)^{-\frac{1}{2}} \approx \frac{1}{3}+\frac{1}{54}x^2+\frac{1}{648}x^4+\ldots$.

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{9-x^2}}\,dx &\approx [\tfrac{1}{3}x+\tfrac{1}{162}x^3+\tfrac{1}{3240}x^4]_0^1 \\ &= (\tfrac{1}{3}+\tfrac{1}{162}+\tfrac{1}{3240})-(0) = \tfrac{367}{1080} \end{aligned}$$

23)

$A = \frac{1}{2}r^2\theta$ gives $x+2 = \frac{1}{2}\times r^2\times 1$. So $r=\sqrt{2x+4}$. $(4+2x)^{\frac{1}{2}} = 4^{\frac{1}{2}}(1+\frac{1}{2}x)^{\frac{1}{2}}$.

$$\begin{aligned} r &\approx 2[1+\tfrac{1}{2}(\tfrac{1}{2}x)+\tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(\tfrac{1}{2}x)^2+\ldots] \\ &\approx 2+\tfrac{1}{2}x-\tfrac{1}{16}x^2+\ldots \end{aligned}$$